IGCSE 0607 Practice Questions — E7.3

Exact trigonometric values

Total Questions: 15 — **Total Marks:** 100 Marks

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  → **Paper 2 (Non-Calculator)** *(15 Qs)*

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1 Without using a calculator, answer the following questions.

(a) Write down the exact value of $\cos 30°$.

..... **[1]**

(b) Show that

$$\frac{1 - \sin 30°}{\tan^2 60°} = \frac{1}{6}$$

[2]

2 Without using a calculator, write down the exact value of each of the following expressions.
You must show all your working.

(a) $\sin 90^\circ + \cos 0^\circ$

..... **[1]**

(b) $\tan 45^\circ \times \sin 30^\circ$

..... **[1]**

(c) $\dfrac{\tan 60^\circ}{\cos 30^\circ}$

..... **[2]**

3 **(a)** Work out the exact value of

$2\sin 60° \times \tan 30°$

..... **[2]**

(b) Find the value of x, where $0° \leq x \leq 90°$, such that

$\sqrt{2}\cos x = \tan 45°$

$x =$ ..... ° **[2]**

4 Without using a calculator, work out the exact value of each of the following.

(a)

$\sin 30^\circ + \cos 60^\circ$

..... **[1]**

(b)

$\tan 60^\circ \times \sin 60^\circ - \cos 0^\circ$

..... **[3]**

5 Do not use a calculator in this question.

(a) Work out the exact value of

(i) $\sin 30° + \cos 60°$,

..... **[2]**

(ii) $\dfrac{\tan 60°}{\cos 30°} - (\sin 45°)^2$.

..... **[3]**

(b) For $0° \leq x \leq 90°$, solve the equation

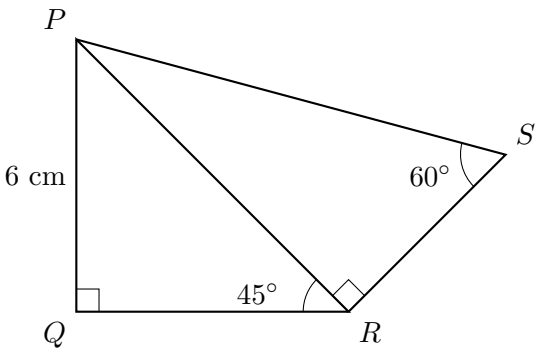
$$2\cos x - \sqrt{3} = 0$$

$x =$ ..... ° **[2]**

(c) Given that $\tan y = \cos 0° - \sin 90°$ and $0° \leq y \leq 90°$, find the value of y.

$y =$ ..... ° **[2]**

6 The diagram shows two right-angled triangles, PQR and PRS.
$PQ = 6$ cm, angle $PQR = 90°$ and angle $PRQ = 45°$.
Angle $PRS = 90°$ and angle $PSR = 60°$.



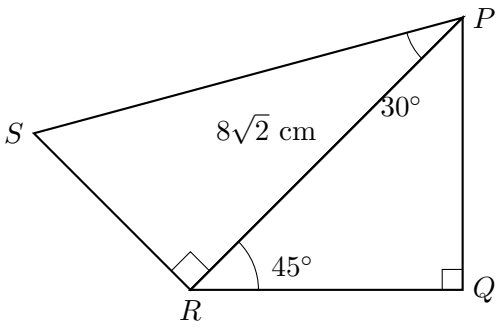
(a) Calculate the exact length of PR.

..... cm **[2]**

(b) Calculate the exact length of PS.
Give your answer in the form $a\sqrt{b}$ where a and b are integers.

..... cm **[3]**

7 The diagram shows two right-angled triangles, PQR and PRS.
Angle $PQR = 90°$ and angle $PRS = 90°$.
Angle $PRQ = 45°$ and angle $SPR = 30°$.
$PR = 8\sqrt{2}$ cm.



NOT TO SCALE

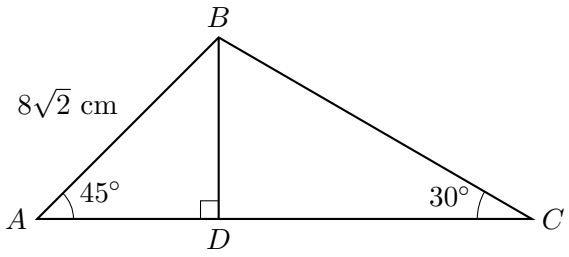
(a) Without using a calculator, show that the exact length of PQ is 8 cm.

[2]

(b) Without using a calculator, work out the exact length of PS.
Give your answer in the form $\dfrac{a\sqrt{b}}{c}$, where a, b and c are integers.

$PS =$ ..... cm **[3]**

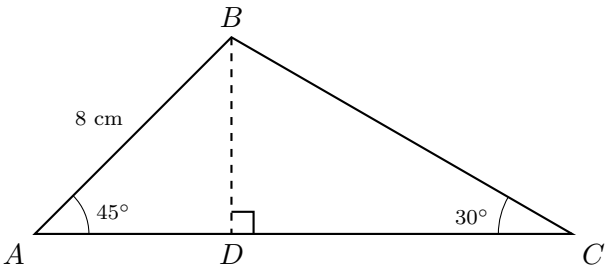
8 The diagram shows a triangle ABC with a point D on the side AC.
BD is perpendicular to AC.
$AB = 8\sqrt{2}$ cm, $\angle BAC = 45°$ and $\angle BCA = 30°$.



Calculate the exact perimeter of triangle ABC.
Give your answer in the form $a + b\sqrt{2} + c\sqrt{3}$ where a, b, and c are integers.

Perimeter = ... cm **[6]**

9 The diagram shows a triangle ABC with $AB = 8$ cm, $\angle BAC = 45°$ and $\angle BCA = 30°$.
BD is perpendicular to AC.



(a) Find the exact length of BD.
Give your answer in its simplest surd form.

$BD =$ ..... *cm* **[2]**

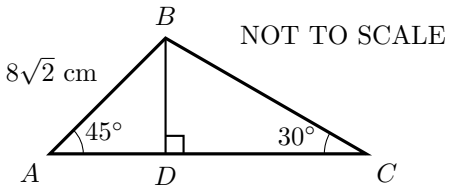
(b) Find the exact length of BC.
Give your answer in its simplest surd form.

$BC =$ ..... *cm* **[2]**

(c) Show that the exact length of AC is $(4\sqrt{2} + 4\sqrt{6})$ cm.

[2]

10 The diagram shows triangle ABC with BD perpendicular to AC, where D lies on AC.
$AB = 8\sqrt{2}$ cm, angle $BAD = 45°$, and angle $BCD = 30°$.



(a) Show that $BD = 8$ cm and $AD = 8$ cm.

[2]

(b) Calculate the exact length of CD, giving your answer in the form $k\sqrt{3}$, where k is an integer.

$CD =$ ..... cm **[2]**

(c) Calculate the exact area of triangle ABC, giving your answer in the form $a + b\sqrt{3}$, where a and b are integers.

Area = ... cm^2 **[3]**

(d) Calculate the exact perimeter of triangle ABC, giving your answer in the form $p + q\sqrt{2} + r\sqrt{3}$, where p, q, and r are integers.

Perimeter = ... cm **[3]**

11 Do not use a calculator for this question.
All working must be clearly shown.

(a) Show that

$$\frac{1 - \tan^2 30^\circ}{1 + \tan^2 30^\circ} = \cos 60^\circ$$

[3]

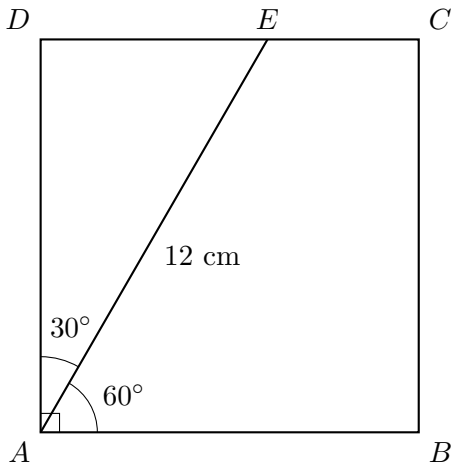
(b) Solve the equation.

$$x \sin 60^\circ = \tan 60^\circ - x \cos 60^\circ$$

Give your answer in the form $a - \sqrt{b}$, where a and b are integers.

$x =$ ..... **[4]**

12 The diagram shows a rectangle $ABCD$.
E is a point on the side CD.
$AE = 12$ cm, angle $DAE = 30°$ and angle $EAB = 60°$.



(a) Show that the exact length of AD is $6\sqrt{3}$ cm.

[2]

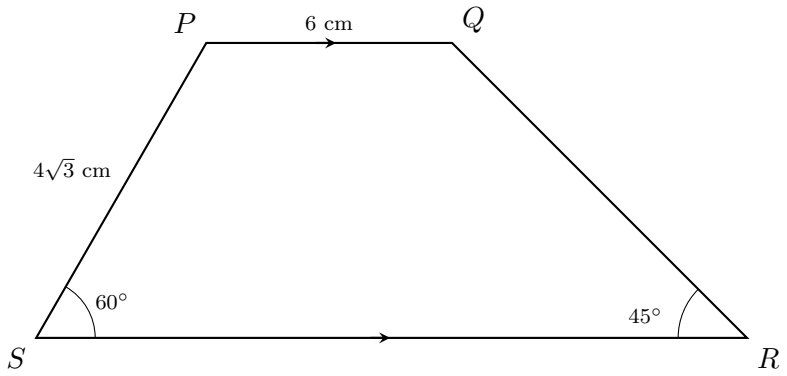
(b) The exact area of the rectangle $ABCD$ is $60\sqrt{3}$ cm^2.
Work out the exact length of CE.

..... cm **[3]**

(c) Find the exact value of $\tan(\angle EBC)$.
Give your answer in the form $\dfrac{a\sqrt{b}}{c}$ where a, b, and c are integers.

..... **[3]**

13 The diagram shows a trapezium $PQRS$ with PQ parallel to SR.
$PQ = 6$ cm, $PS = 4\sqrt{3}$ cm, $\angle PSR = 60°$ and $\angle QRS = 45°$.



(a) Find the exact perpendicular height of the trapezium $PQRS$.

Height = ... cm **[2]**

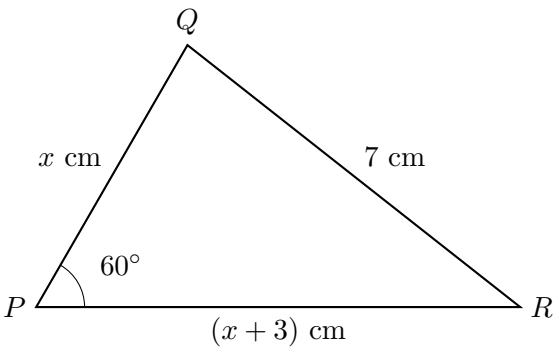
(b) Calculate the exact area of the trapezium $PQRS$.
Give your answer in the form $a + b\sqrt{3}$, where a and b are integers.

Area = ... cm^2 **[4]**

(c) Show that the exact perimeter of the trapezium $PQRS$ is $(18 + 6\sqrt{3} + 6\sqrt{2})$ cm.

[2]

14 The diagram shows a triangle PQR.
$PQ = x$ cm, $PR = (x + 3)$ cm and $QR = 7$ cm.
Angle $QPR = 60°$.



(a) Show that $x^2 + 3x - 40 = 0$.

[4]

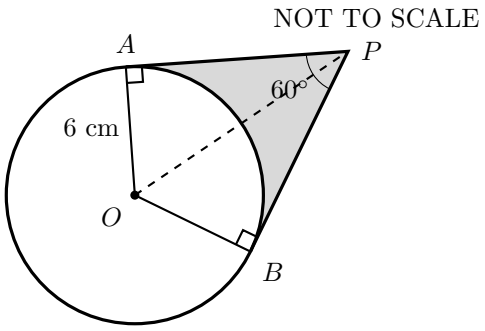
(b) Solve the equation to find the value of x, and hence find the perimeter of triangle PQR.

Perimeter = ... cm **[3]**

(c) Find the exact area of triangle PQR.

Area = ... cm^2 **[3]**

15 The diagram shows a circle, centre O, of radius 6 cm. From an external point P, tangents PA and PB are drawn to touch the circle at A and B respectively, such that angle $APB = 60°$.
The region bounded by PA, PB, and the minor arc AB is shaded.



(a) Find the exact length of the tangent PA, giving your answer in the form $k\sqrt{3}$, where k is an integer.

$PA =$ ..... cm **[3]**

(b) Calculate the exact area of the shaded region, giving your answer in the form $a\sqrt{3} - b\pi$, where a and b are integers.

Area = ... cm^2 **[4]**

(c) Calculate

(i) the exact length of the chord AB, giving your answer in the form $m\sqrt{3}$, where m is an integer,

$AB =$ ..... cm **[2]**

(ii) the exact shortest distance from point P to the chord AB.

Distance = ... cm **[2]**